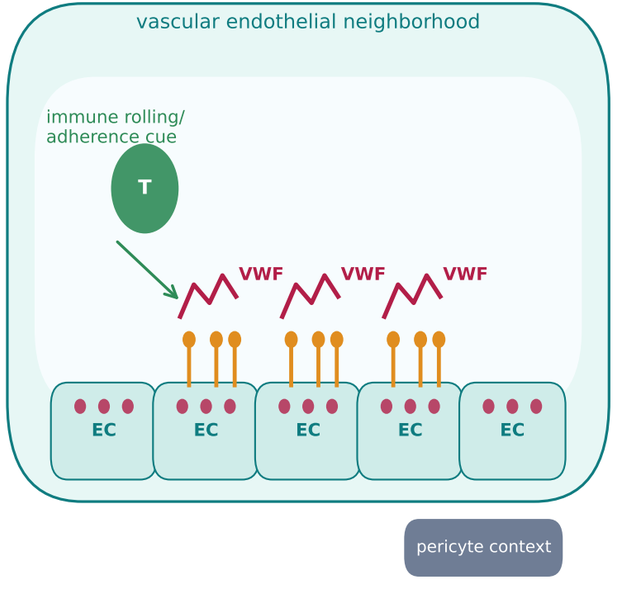
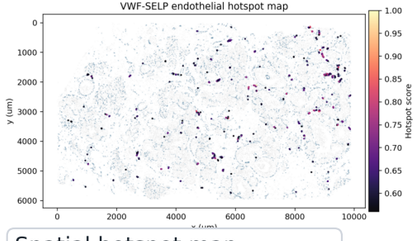
VWF-SELP marks an endothelial activation niche

A topology-supported pyXenium top hit linking endothelial identity, WPB biology, spatial hotspots, and vascular tissue ecology.

A Endothelial activation model

VWF + SELP/P-selectin = WPB / adhesion state





Spatial hotspot map
433 endothelial hotspot cells

pyXenium rank 1

LR_score 0.791; contact-aware topology keeps VWF-SELP as the top endothelial axis.

WTA specificity

VWF and SELP peak in endothelial cells with PECAM1/EMCN/KDR support.

Neighborhood context

Hotspot ECs are enriched near T-cell and pericyte neighborhoods.

Contour ecology

S1-S5 analysis places vascular markers in S3-enriched tissue regions.

Cross-method support

CellPhoneDB and LARIS recover exact VWF-SELP endothelial-endothelial support.

Interpretation

RNA/spatial evidence for WPB-associated adhesion state; protein validation needed.

Biological model: activated endothelial neighborhoods expose a VWF/P-selectin-like adhesion program that may organize immune and pericyte-adjacent vascular niches.

Caveat: RNA + spatial evidence; not direct proof of protein release or thrombosis.